

Android development in IntelliJ IDEA: For Android development, add the Android support plugins bundle in IntelliJ IDEA from the IntelliJ plugin repository.

The Android development plugin bundles: From the bundles, various tools are available for particular tasks:

1. For creating an Android project build: Gradle based build system.
2. Facility for working with the layout, manifest, resource and other XML files: Smart XML editor.
3. Specific code inspections for Android use: Android-specific code inspections including those integrated with Android Lint inspections.
4. To build the design for your application without changing the XML files manually: Graphics editor.
5. For the database: Access to the SQLite database.

These are the basic plugin bundles for developing the Android Project.

Stable release	2018.2.4 Build: 182.4505.22
Licence	Community edition: <i>Apache 2.0 License</i>
Website	www.jetbrains.com/idea/
Written in	Java
OSs supported	Windows, MacOS, Linux
Main features	Supported communications languages: Java, Groovy, Perl, Rust, Scala, XML/XSL, Kotlin, Python, Lua, Haskell, etc Technology and framework: Android, Ant, Gradle, Junit, Maven, TestNG, NodeJS, JSP, Spring, etc Version controller: CVS, Git, GitHub, SVN (Subversion), Mercurial, Team Foundation Server (TFS) Database support: Oracle, Microsoft SQL Server, MySQL and PostgreSQL

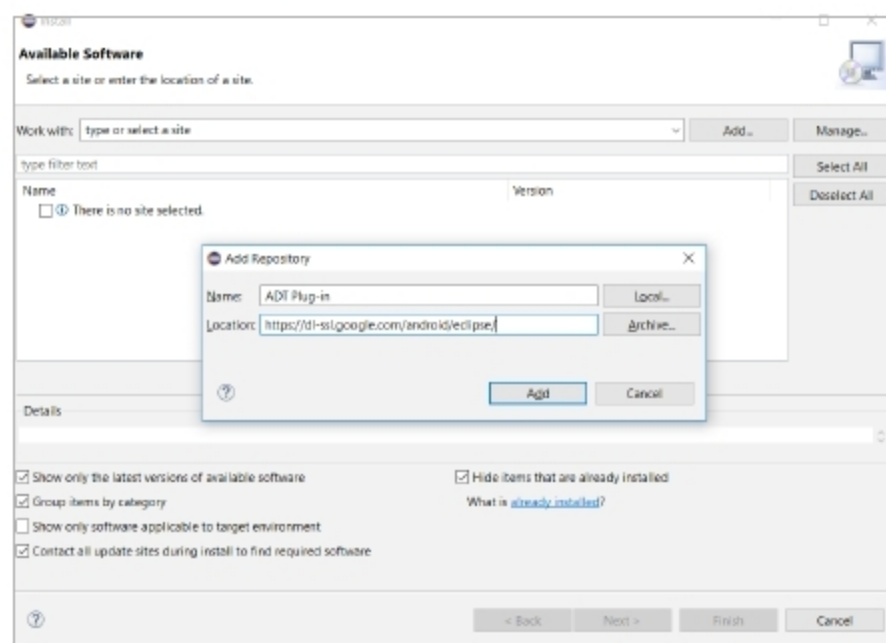


Figure 10: ADT plugin in Eclipse

Eclipse

Eclipse is one of the most popular IDEs among developers. Download and install it from <https://www.eclipse.org/downloads/>.

To set the Android Development Tool plug-in for Eclipse, open Eclipse and then choose *Help > Software Updates > Install New Software*. Add Repository and provide values as given.

Click on *Add*.

Check the available development tools and click on *Next*. Click on *File > New > Others > Select Android* and start working.

Stable release	4.9 (2018-09 R)
Licence	Eclipse Public
Website	https://www.eclipse.org/
Written in	Java and C
Supported on	Windows, Linux, MacOS and Solaris
Main features	Git flow support. It provides a set of modelling tools.

Continuous integration

Continuous integration (CI) is one of the popular practices of DevOps. Developers commit code daily in the code repository such as Git or SVN, which triggers a process that includes static code analysis, compilation, unit test execution and package creation. In case of Android development, it creates an APK file.

If the process is followed correctly and if it becomes a part of the work culture, continuous integration helps to identify issues early on so that they can be solved easily. Fast detection of failures results in instant and constant feedback and hence the quality of the APK files improves even before these are made available to users.

There are many CI tools in the market. Jenkins is one of the most popular open source tools to implement CI for Android projects.

Jenkins

Jenkins is no longer a CI server since Jenkins 2.0, and it is

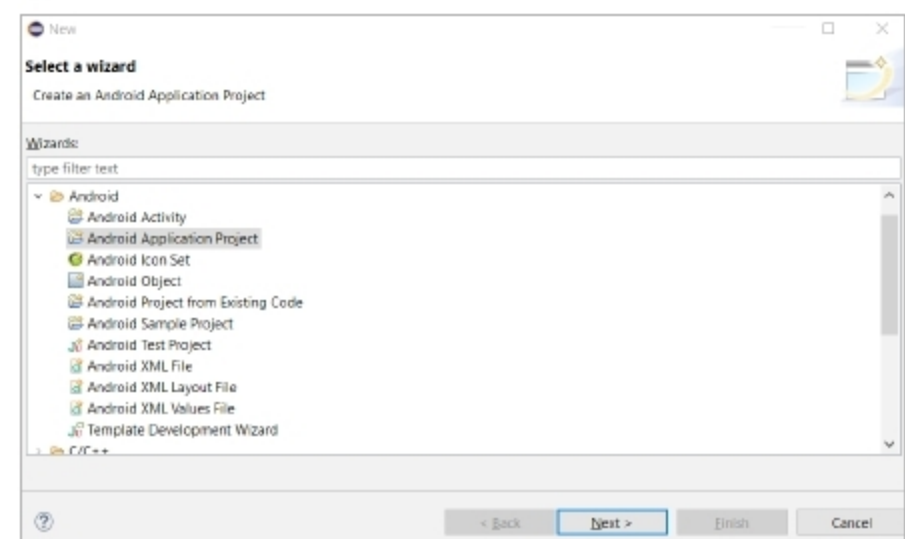


Figure 11: Android Application Project Wizard